A Robust Oriented Filter-Based Matching Method for Multisource, Multitemporal Remote Sensing Images

Zhongli Fan, Mi Wang, Yingdong Pi, Yuxuan Liu, and Huiwei Jiang

***Abstract*— The accurate matching of multisource, multitemporal remote sensing images is challenging because of significant nonlinear intensity differences (NIDs) and severe geometric distortions. To address these problems, we developed a robust image matching method: oriented filter-based matching (OFM). OFM is insensitive to NIDs while exhibiting scale and rotational invariance. First, salient feature points with multiscale attributes were detected in the Gaussian-scale space of the input images. Then, the images were convoluted using multioriented filters, and unified feature maps were constructed by the extraction of orientation indices using effective data pooling operations. The constructed feature maps were highly resistant to NIDs. Five filters were integrated into the OFM framework to investigate their applicabilities in different application scenarios. Next, a novel rotation-invariant feature descriptor was constructed, using a dominant direction determination approach and a descriptor-grouping strategy. The dominant direction determination approach enables accurate dominant direction estimation, whereas the descriptor-grouping strategy improves the stability of the method under different rotational angles. Finally, brute-force matching was implemented to obtain initial matches; an improved mismatch elimination method was used to identify reliable putative matches. To evaluate the performance of OFM, we created a large dataset comprising 4427 pairs of multitemporal optical–optical, optical–synthetic aperture radar (SAR), optical–infrared, and optical–depth images. OFM outperformed state-of-the-art methods in terms of a number of correct matches (NCM), recall, inlier ratio, root mean square error (RMSE), and success rate (SR). Our implementation is publicly available at https://github.com/Zhongli-Fan/OFM.**

***Index Terms*— Image matching, multisource images, multitemporal images, nonlinear intensity differences (NIDs), oriented filers.**

I. Introduction

IMAGE matching is a fundamental prerequisite for many applications in the fields of remote sensing, such as image stitching [1], mosaicking [2], fusion [3], and change detection [4]. The accuracy and effectiveness of these applications are significantly influenced by image matching results; in many instances, high-precision image matching is desired. Rapid advances in sensor and remote sensing technologies have led to increasing attention toward the joint processing of multisource remote sensing data in both academia and industry, and the demand for co-registration of multisource remote sensing images (e.g., optical, synthetic aperture radar (SAR), infrared, and LiDAR depth, etc.) has substantially grown. Improved multitemporal image matching is also needed to meet the requirements of applications related to remote sensing time-series analysis [5].

In the past few decades, image matching techniques have been developed significantly. Most of the current methods (e.g., scale-invariant feature transform (SIFT) [6], oriented fast and rotated brief (ORB) [7], and cross correlation, etc.) are designed for homo-source images and they can only handle linear intensity differences. However, for multisource, multitemporal images, differences in the imaging mechanism and imaging time result in significant nonlinear intensity differences (NIDs) that are further exacerbated by variation in illumination and contrast conditions. Various types of images are also subject to noise interference, such as strong speckle noise on SAR images, and clouds and fog on optical and infrared images. Although global geometric distortions between images can be eliminated using direct georeferencing, severe local geometric distortions can persist because of significant differences in acquisition principles and viewing perspectives (e.g., optical satellites usually adopt down-view imaging, whereas SAR satellites adopt side-view imaging). For example, consider the pair of optical–SAR images shown in Fig. 1(a); these images have significant NIDs and local geometric distortions between them, particularly for tall buildings. Speckle noise interference is another serious obstacle in SAR images. Fig. 1(b) shows a representative pair of multitemporal optical images acquired 15 years apart, during which regional contents substantially changed and the available information was significantly reduced; seasonal changes and fog disturbances further exacerbated differences between the images. These types of factors can hinder reliable image matching, increasing the difficulty of joint applications for multisource, multitemporal images.

In this study, to address these difficulties, we designed a novel image matching method that is robust to NIDs and has both rotational and scale invariance: oriented filter-based matching (OFM). In OFM, salient feature points are first

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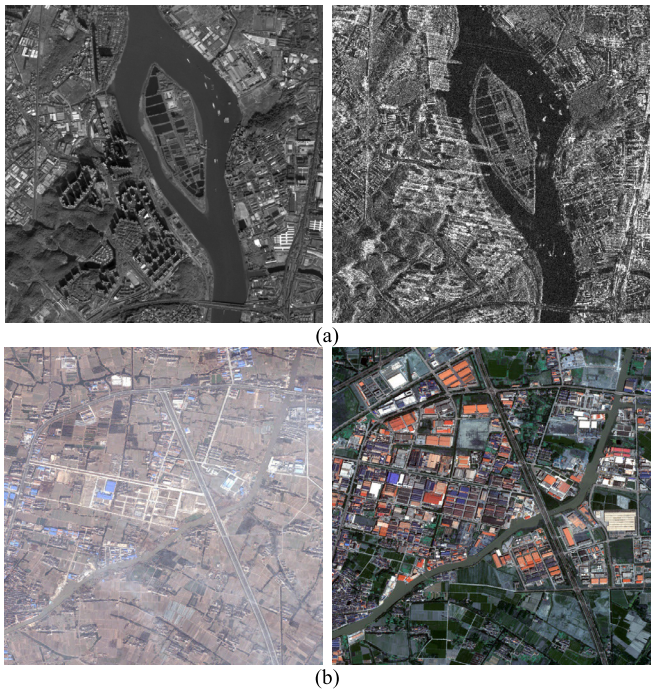


Fig. 1. Two pairs of typical of multisource, multitemporal images. (a) Pair of optical–SAR images. (b) Pair of multitemporal optical images.

extracted based on a phase congruency (PC) model [8]; the resulting features are given scale properties through the construction of a Gaussian-scale space for each image. To resist significant NIDs, an orientation index map (OIM) is constructed to generate a unified image pattern based on multidirectional image convolution and an effective data pooling operation. Then, a novel rotation-invariant feature descriptor is designed using a stable dominant direction determination scheme. Based on the directionality of the convolution, a descriptor-grouping strategy is used to improve the rotational resistance of OFM. Finally, the initial matches are obtained by a brute-force matching procedure; reliable putative matches are determined by a random sample consensus (RANSAC) algorithm [9] and a match set iterative optimization (MSIO) strategy. It is worth highlighting that five oriented filters are integrated into the OFM framework based on the consideration of various application scenarios: first- and second-order Gaussian steerable filters (G1StF and G2StF), approximate Hilbert transform of G2StF (H2StF), and Gabor and log-Gabor oriented filters (GoF and LGoF).

The main contributions of this study are as follows.

1) OFM was designed as an NID-insensitive matching method for multisource, multitemporal remote sensing images with the capacity to obtain stable, reliable matched point pairs in a rotation- and scale-invariant manner.
2) Five different oriented filters were integrated into the OFM framework, and their performance in different application scenarios was investigated and analyzed in detail.
3) Extensive experimental results demonstrated the superiority of our OFM approach, in comparison with several state-of-the-art methods, when using a large dataset that comprised 4427 image pairs.

The remainder of this article is organized as follows. Section II presents related works concerning the topic of image matching. Section III introduces the proposed OFM method. Section IV presents the experimental results and analyses. Finally, Section V provides our conclusion.

II. Related Works

According to the central ideas of current multisource, multitemporal remote sensing image matching methods, they can be broadly classified as area-based methods and feature-based methods [10], [11].

A. Area-Based Methods

Area-based methods are commonly used in the fields of remote sensing to achieve image matching through a template matching strategy. Specifically, they establish template windows on the images and calculate similarity by sliding the windows in a pixelwise manner to determine the matching position. Conventional area-based methods are distinguished by the design of similarity measures; well-known methods include normalized cross correlation (NCC) [12], which is robust to linear intensity variation. However, NCC is highly sensitive to NIDs because it uses image intensity as a direct input for the matching process. Constantini et al. [13] proposed a two-step registration method for optical and SAR images. They first eliminated global image displacements between images by a bundle adjustment procedure. After that, the registration is achieved through a matching step based on mutual information (MI) algorithm [14] that is well resistance to NIDs. However, MI is only applicable to local optimal solutions with small offsets, and it experiences difficulty in finding global optimal solutions because of its high runtime complexity. To resist NIDs, recent methods have focused on template designs, which is a more effective approach than the design of similarity measures. Based on the histogram of oriented gradients (HOG) [15] and the PC model, Ye et al. [16] proposed an improved image template, the histogram of oriented PC (HOPC), which is constructed by extracting structural information from an image that is stable across multisource images. Based on HOPC, Xiong et al. [17] proposed a rank-based local self-similarity (RLSS) template for optical and SAR image registration. RLSS effectively describes the shape properties of an image. Although HOPC and RLSS can be used to match multisource images, they are both computed on a sparse sampling grid and have limited capacity to represent detailed structural information. Therefore, Ye et al. [18] proposed a densely structured image template, channel feature of oriented gradients (CFOG), which efficiently captures image structural features and achieves rapid matching using a fast Fourier transform technique. With inspiration from the CFOG template, Fan et al. [19] proposed an angle-weighted oriented gradient template, which uses the gradient direction to weight the assigned gradient magnitude, thereby improving its performance relative to CFOG. For the registration of SAR and optical Images, Li et al. [20]

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designed a densely 3-D image descriptor based on multiscale Gabor filters. This descriptor is able to represent the image structure well. And they achieved fast image matching using an improved similarity measure based on sum of squared difference.

B. Feature-Based Methods

Feature-based methods achieve image matching using notable and invariant features. These methods detect salient features in images and generate distinguishable feature description information. Based on comparisons of distances between descriptors, matching point pairs are determined and collected. Rasmy et al. [21] proposed a registration method for multitemporal aerial images. They first used the Harris detector to extract feature points on the reference and sensed images, then used Fourier phase matching to search for corresponding points, and finally achieved accurate registration using two optimization algorithms. However, this method cannot handle images with rotation or scale differences. The difference of Gaussians (DoG) that used in SIFT algorithm [6] is one of the most popular feature detectors used for image matching. Although SIFT is rotation and scale invariant, it is sensitive to NIDs. Some improved versions of SIFT have been proposed such as position, scale, orientation-SIFT (PSO-SIFT) [22], and optical-to-SAR-SIFT (OS-SIFT) [23]. PSO-SIFT uses second-order gradients to construct descriptors and statistics regarding translational, rotational, and scale differences between images for the estimation of an optimal geometric transformation model. It has demonstrated good performance for multitemporal optical–optical images and optical–infrared images. For the registration of optical–SAR image pairs, OS-SIFT computes their gradients using multiscale Sobel and ratio of exponentially weighted averages operators, respectively. However, OS-SIFT can only obtain a small number of matches, which hinders its capacity to ensure the matching accuracy.

To address the problem of multisource image matching, advanced methods have been developed that involve constructing feature maps based on the orientation index, rather than gradients. Using multiscale, multioriented log-Gabor filters, Aguilera et al. [24] proposed a log-Gabor histogram descriptor (LGHD) algorithm to effectively match visible and long-wave infrared image pairs. However, LGHD uses the original features from accelerated segment test (FAST) [25] detector for feature extraction. FAST is among the best corner detectors available, considering its capacity to extract large number of features within very short periods, but it still has some limitations, such as large differences in the responses of extracted features across image regions, which negatively impact feature distribution, and a significant reduction in the number of extracted features when the input image has poor illumination and contrast conditions. Moreover, LGHD has incomplete feature dimensionality reduction, which leads to high feature-vector dimensionality and low computational efficiency. Similar to LGHD, Li et al. [26] proposed a radiation variation-insensitive feature transform (RIFT) algorithm based on identical log-Gabor filters. RIFT reduces the dimensionality of the descriptors by summing the convolution results over multiple scales and constructs an end-to-end loop structure to achieve rotational invariance. Adhering to this logic, Xiong et al. [27] proposed a matching method based on self-similarity features. They designed a novel descriptor, oriented self-similarity (OSS), and achieved reliable matching results. Xiong et al. [28] further proposed an adjacent self-similarity (ASS) algorithm for the alignment of optical and SAR images. ASS uses local self-similarity descriptors to generate feature maps and achieve rotational invariance via direction voting and descriptor rotation strategies. By converting multisource images into the similar modalities, Li et al. [29] proposed a locally normalized image feature transform (LNIFT) method. LNIFT achieves rotational invariance by using an improved ORB detector and a simplified HOG descriptor.

Artificial intelligence technology has undergone rapid development in recent years. Notably, deep learning has been introduced into fields such as image feature extraction and matching. Yu et al. [30] investigated the application of multisource image matching for urban environments using road features. The method involved the detection of unstructured feature points on semantic road features that were extracted by the semantic segmentation model CRESIv2, followed by the use of an entropy-based descriptor for feature description of the local road skeleton around the feature points. Hughes et al. [31] proposed a three-step registration method for optical and SAR images, with each step encoded by a deep neural network. The first neural network detects salient feature points, the second neural network generates a correspondence heatmap, and the third neural network extracts putative matches. Quan et al. [32] used a self-distillation neural network for feature extraction and matching of optical and SAR images. The method involved initially designing a partially shared feature extraction network, then enhancing similarity information between images through the introduction of a self-distillation feature learning approach, and finally constructing a task reconstruction learning algorithm to retain distinguishing information. Zhang [33] employed a deep learning framework to generate pseudo-optical images from SAR images, and then achieved the matching of these translated images using a traditional method under the same image model.

However, current deep learning methods suffer from several problems. First, there is a lack of a multisource remote sensing dataset large enough to train deep neural networks. Second, they are less generalizable across different types of multisource images. Finally, they bring challenges to the computer configurations. For practical applications, deep learning methods may need to shift toward self-supervised learning and lightweight network structures.

In summary, area-based methods are effective for the matching of multisource, multitemporal images; however, they have two main limitations that hinder their generality to various application scenarios. First, they function within a small search area (typically a radius of $\leq$100 pixels) and their performance substantially decreases if the initial predicted position significantly deviates from the correct match. Second, they can only match images with translational differences; they

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exhibit minimal resistance to large geometric differences such as rotational and scale variation.

In contrast, feature-based methods are robust to NIDs and large geometric distortions. Nevertheless, most current methods cannot effectively manage scale differences and their performance is highly unstable under different image rotational angles; moreover, they exhibit limited generalizability toward different types of multisource image data because their applicabilities to various convolution kernels have not been extensively characterized.

The proposed OFM method is a feature-based method, but not a deep learning method. To achieve the robust matching of multisource, multitemporal images, OFM mainly relies on four key components.

1) The use of multiscale salient features as a basis for generating scale invariance.
2) A unified OIM to resist significant NIDs.
3) A novel rotationally invariant feature descriptor to achieve robust image matching.
4) An improved mismatch elimination approach with an MSIO strategy.

III. Methodology

The OFM method mainly includes three steps, as illustrated in Fig. 2. In the first step, Gaussian-scale spaces are constructed for reference and sensed images; a set of uniformly distributed feature points is extracted from different scale layers of a Gaussian pyramid using PC-FAST detectors [10]. In the second step, a unique feature vector is captured for each feature point to identify correspondence. The essential aspect of this step is a novel feature descriptor that is generated based on a class of multioriented filters (e.g., steerable, Gabor, or log-Gabor filters). A novel dominant direction determination approach is used to achieve descriptor rotational invariance. In the last step, brute-force matching is performed to obtain a set of initial matches. Finally, a mismatch elimination method is designed by combining a RANSAC algorithm and an MSIO strategy to ensure the accuracy and reliability of the matching results. The OFM method is described in detail in Sections III-A–III-C.

A. Feature Extraction

According to our previous research on the PC model [10], PC-FAST significantly improves feature distribution and increases the number of features in images captured by different sensors. The PC model is a well-known image edge extraction model with illumination and contrast invariance. A PC maximum moment map, which contains more detailed edges than the original PC map, can be generated by applying moment analysis functions to the PC map in a different direction. The implementation details of PC-FAST can be found in [10]. The main steps of the multiscale PC-FAST detector used in this study are the construction of a Gaussian-scale space of the input image, followed by the generation of a PC maximum moment map for each layer and the performance of FAST detection on these maps to extract feature points. Multiscale FAST and PC-FAST feature extraction results for an example image [see Fig. 3(a)] are shown in Fig. 3(b) and (c), respectively. A comparison of these results with the results of the original FAST detector under identical parameter settings shows that PC-FAST retains more features and exhibits a more uniform spatial distribution.

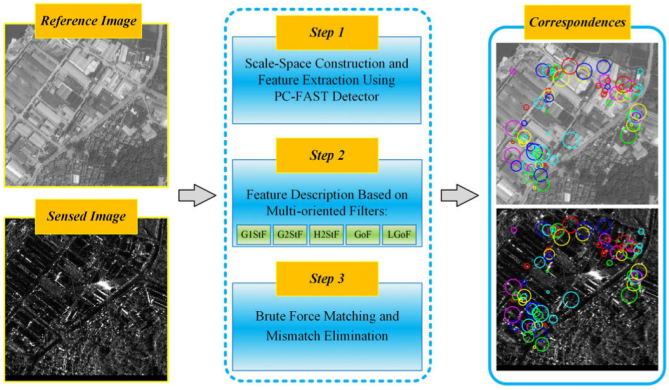
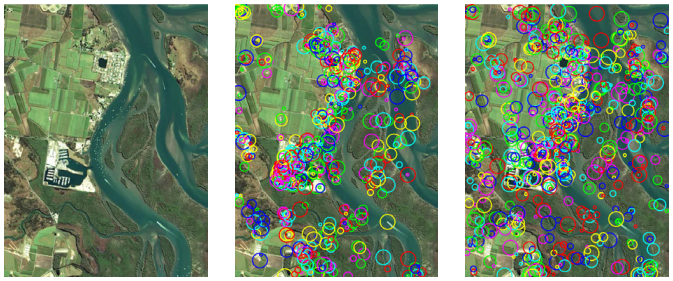


Fig. 2. Main steps of the proposed OFM method.



(a) (b) (c)

Fig. 3. Feature extraction results. (a) Example image. (b) Multiscale FAST features. (c) Multiscale PC-FAST features. The colors and radii of the circles indicate different feature points and scales.

B. Feature Description

To match features, feature description vectors are computed on local image regions that are centered on target feature points. Robust image feature description based on multioriented filters has great potential, considering that the multioriented convolution results provide more detailed image information and reflect directional differences in feature responses. In this section, we propose a novel rotationally invariant descriptor based on multioriented filters, using the steerable filter H2StF as an example.

A filter of arbitrary orientation is considered "steerable" if it can be synthesized as a linear combination of a set of specific basis filters [34]. As one of the simplest steerable filters, H2StF can be formed using four third-order Gaussian derivative operators as basis filters. The 2-D Gaussian function represented in Cartesian coordinates (x, y) can be expressed as follows:

$$G(x, y) = \frac{1}{2\pi\sigma^2} e^{\frac{-(x^2+y^2)}{2\sigma^2}} \tag{1}$$

where $G(x, y)$ is the Gaussian function and σ is the standard deviation of $G(x, y)$.

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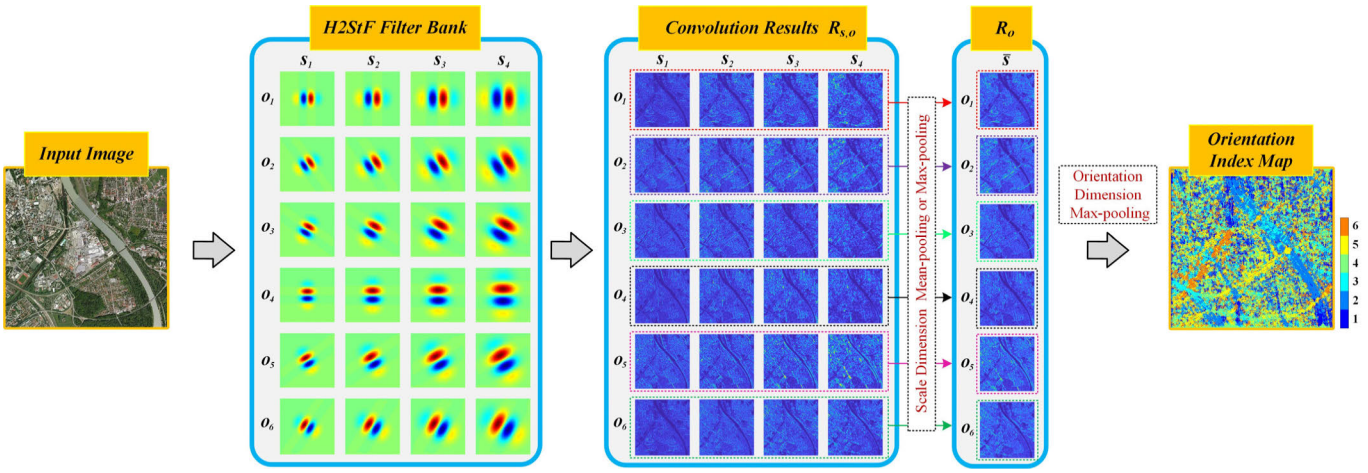


Fig. 4. Construction process of an OIM using H2StF.

The third-order Gaussian derivative operators of $G(x, y)$ are used as basis filters for H2StF, as follows:

$$\begin{cases} H^a_{2,\sigma}(x,y) = \dfrac{\partial^3 G}{\partial x^3} = \dfrac{1}{2\pi\sigma^6}\left(3x - \dfrac{x^3}{\sigma^2}\right)e^{\frac{-(x^2+y^2)}{2\sigma^2}} \\ H^b_{2,\sigma}(x,y) = \dfrac{\partial^3 G}{\partial x^2 \partial y} = \dfrac{1}{2\pi\sigma^6}\left(1 - \dfrac{x^2}{\sigma^2}\right)ye^{\frac{-(x^2+y^2)}{2\sigma^2}} \\ H^c_{2,\sigma}(x,y) = \dfrac{\partial^3 G}{\partial y^2 \partial x} = \dfrac{1}{2\pi\sigma^6}\left(1 - \dfrac{y^2}{\sigma^2}\right)xe^{\frac{-(x^2+y^2)}{2\sigma^2}} \\ H^d_{2,\sigma}(x,y) = \dfrac{\partial^3 G}{\partial y^3} = \dfrac{1}{2\pi\sigma^6}\left(3y - \dfrac{y^3}{\sigma^2}\right)e^{\frac{-(x^2+y^2)}{2\sigma^2}} \end{cases} \tag{2}$$

where a, b, c, and d are used to distinguish the basis filters.

The H2StF at arbitrary orientation θ can be synthesized using a linear combination of $H^a_{2,\sigma}$, $H^b_{2,\sigma}$, $H^c_{2,\sigma}$, and $H^d_{2,\sigma}$, as follows:

$$\begin{cases} H^\theta_{2,\sigma}(x,y) = \displaystyle\sum_{i=1}^{4} k_{L_i}(\theta) H^{L_i}_{2,\sigma}(x,y), \;\; L_i = [a,b,c,d] \\ k_a(\theta) = \cos^3(\theta) \\ k_b(\theta) = -3\cos^2(\theta)\sin(\theta) \\ k_c(\theta) = -3\cos(\theta)\sin^2(\theta) \\ k_d(\theta) = \sin^3(\theta) \end{cases} \tag{3}$$

where $k_a(\theta)$, $k_b(\theta)$, $k_c(\theta)$, and $k_d(\theta)$ are interpolation functions for fitting to the Hilbert transform of the second derivative of the Gaussian function.

In addition to steerable filters like H2StF, any wavelet-based multioriented filters such as GoF and LGoF can also be used for multidirectional image convolution. These multioriented filters mentioned above are generated in a manner based directly or indirectly on the Gaussian function. Therefore, adaptive control of these filters can be achieved by adjusting the directional angle settings and standard deviations of the Gaussian function to create a multioriented, multiscale filter bank.

Similar to LGHD [24], the descriptor proposed in this study is constructed based on multidirectional image convolution. LGHD explored feature representation based on orientation indicators provided by LGoF to improve image similarity. However, LGHD has two main limitations: incomplete feature dimensionality reduction, and a lack of scale and rotational invariance. Incomplete feature dimensionality reduction increases computational cost and reduces the distinguishability of descriptors, whereas the absence of scale and rotational invariance severely limits the applicability of matching algorithms. Therefore, the proposed descriptor uses a pooling operation inspired from deep learning field to reduce feature dimensionality; it uses a novel dominant direction estimation approach to achieve rotational invariance.

In contrast to conventional descriptors that use an image intensity or gradient map to compute feature vectors, descriptors based on multidirectional convolution use an OIM as input data. The construction of the proposed OIM is shown in Fig. 4.

As shown in the figure, the input image $I(x, y)$ is first convolved using a multioriented, multiscale H2StF filter bank to capture image structure features in various orientations and scales. The number of orientations in the filter bank is N_o, and the number of scales is N_s. Through this process, we calculate the set of convolution results $R_{s,o}(x, y)$ as follows:

$$\begin{cases} R_{s_i,o_j}(x,y) = \text{abs}\left[H^{\theta_j}_{2,\sigma_i} * I(x,y)\right] \\ \theta_j = \dfrac{\pi(j-1)}{N_o}, \quad j \in [1, N_o] \end{cases} \tag{4}$$

where the subscripts s_i and o_j correspond to the Gaussian standard deviation σ_i and orientation angle θ_j, respectively; the asterisk $*$ indicates the convolution calculation; and the function abs$[\cdot]$ is used to calculate the absolute value. H2StF is an odd-symmetric filter with respect to its orientation. When the filter $H^{\theta_j}_{2,\sigma_i}$ is rotated by π radians, the equation $H^{\theta_j+\pi}_{2,\sigma_i}(x, y) = -H^{\theta_j}_{2,\sigma_i}(x, y)$ is applicable to any position (x, y) in its domain, and the equation $R^{\theta_j+\pi}_{2,\sigma_i}(x, y) = -R^{\theta_j}_{2,\sigma_i}(x, y)$ is applicable to the convolution results. Accordingly, identical convolution results of filters $H^{\theta_j}_{2,\sigma_i}$ and $H^{\theta_j+\pi}_{2,\sigma_i}$ can be obtained by performing the absolute calculation shown in (4), with the directional angle θ of the filter set at $[0,\pi)$ rather than $[0, 2\pi)$. Thus, the convolution results of an image and its π-radians rotation are identical.

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After the convolution calculation, $N_o \times N_s$ convolutional images are obtained, resulting in a very large amount of data that includes many invalid features. Therefore, feature dimensionality reduction must be performed to highlight essential image feature information. Here, we introduce an effective pooling operation to achieve feature dimensionality reduction of $R_{s,o}(x, y)$. Classical pooling operations are performed on the image plane, resulting in the reduction of image resolution. For image matching, such reduction is undesirable because it will decrease matching accuracy; therefore, the pooling operation is performed on the two new feature dimensions (scale and orientation). Mean pooling and max pooling are the two most commonly used pooling approaches. Generally, mean pooling makes features smoother, and max pooling makes features more prominent. Feature dimensionality reduction can be achieved by using mean pooling to smooth features and attenuate the influence of invalid features, or by using max pooling to highlight salient features. As shown in Fig. 4, convolution results in the scale dimension are first pooled in each fixed orientation. In this step, the pooling operation can be either mean or max pooling. Pooling operations in the scale dimension are expressed as follows:

$$\begin{cases} R_{o_j}(x, y) = \dfrac{1}{N_s} \displaystyle\sum_{i=1}^{N_s} R_{s_i, o_j}(x, y) \\ \text{or} \\ R_{o_j}(x, y) = \max_s \left[R_{s_i, o_j}(x, y) \right] \end{cases} \tag{5}$$

where the first and second formulas represent mean and max pooling operations in the scale dimension, respectively, and the function $\max_s[\cdot]$ is used to obtain the largest feature value. Feature information for convolutional images at different scales is integrated into a single image, and the scale dimension is simultaneously eliminated.

Subsequently, max pooling is performed in the orientation dimension to further reduce the feature dimension. The essential aspect of this step is the acquisition of an orientation index that corresponds to the largest feature value, rather than the largest feature value itself. An OIM can be constructed by collecting the orientation indices j at each location (x, y) of the convolution result $R_{o_j}(x, y)$. Mean pooling cannot be used in this step because the mean of the feature values in each orientation has no corresponding orientation index. The max pooling operation in the orientation dimension is performed as follows:

$$\text{OIM}(x, y) = \max_j \left[R_{o_j}(x, y) \right], \quad j \in [1, N_o] \tag{6}$$

where $\text{OIM}(x, y)$ is the OIM result after the entire feature dimensionality reduction process, and the function $\max_j[\cdot]$ is used to obtain the orientation index of the largest feature value. After orientation max pooling, the orientation dimension is eliminated and the OIM is obtained. OIMs for a typical optical–SAR image pair [see Fig. 5(a)] generated by G1StF, G2StF, H2StF, GoF, and LGoF, respectively, are shown in Fig. 5(b)–(f). The similarity of the OIMs is higher than the similarity of the original images.

The orientation index that corresponds to the largest feature value, rather than the largest feature value itself, is used to build the feature map because this approach overcomes the significant NIDs between target images. The multidirectional convolution operation captures image structural properties, but it cannot provide a unified measure of feature values for each image type. However, a single ground object, which does not change between acquisition dates, should have identical or similar structural properties, although it may be imaged by different sensors; the orientation index corresponding to the largest feature value should be stable in different images, although the largest feature value itself is unpredictable. Therefore, the orientation index can be used as a compromise unified measure to construct the feature map.

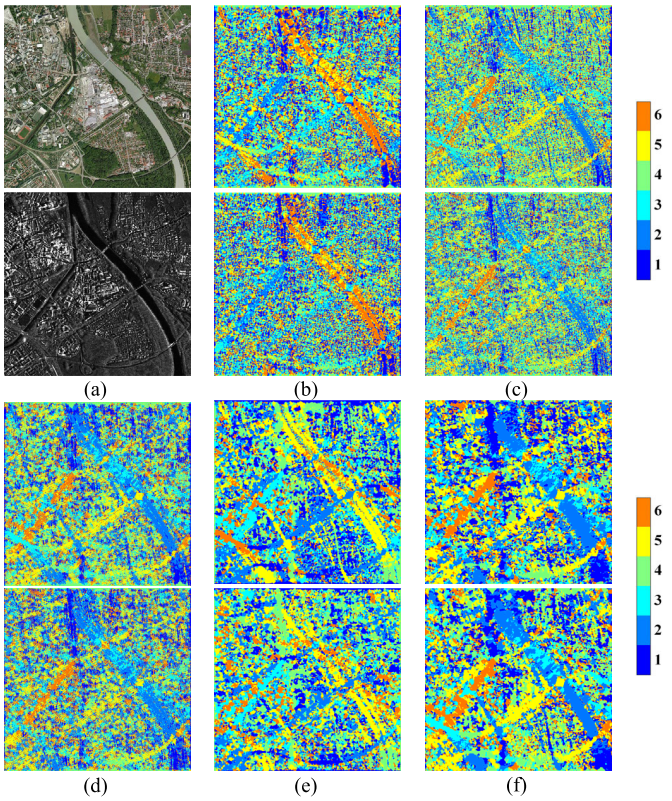


Fig. 5. OIMs of a typical optical–SAR image pair generated by different multioriented filters. (a) Original optical–SAR image pair. (b)–(f) OIMs generated by the G1StF, G2tF, H2StF, GoF, and LGoF, respectively.

However, the OIMs are not rotational invariant because the orientation angle θ of each filter in the filter bank is fixed, such that when the image has rotational changes, the orientation index of the largest feature value becomes unpredictable. Nevertheless, a change in rotation does not alter the structural information within the image. Accordingly, although the value of the OIM changes with image rotation, its internal structure and value distribution remain stable, providing an opportunity for the descriptor to achieve self-adaptation of the image rotational angle. Therefore, we further propose a novel rotational invariant descriptor based on local consistency of the OIM, which is constructed as illustrated in Fig. 6.

The proposed descriptor is generated based on local image regions that are centered on extracted feature points, and all local regions are set as circular regions with a constant radius to achieve the scale invariance. Because a feature point

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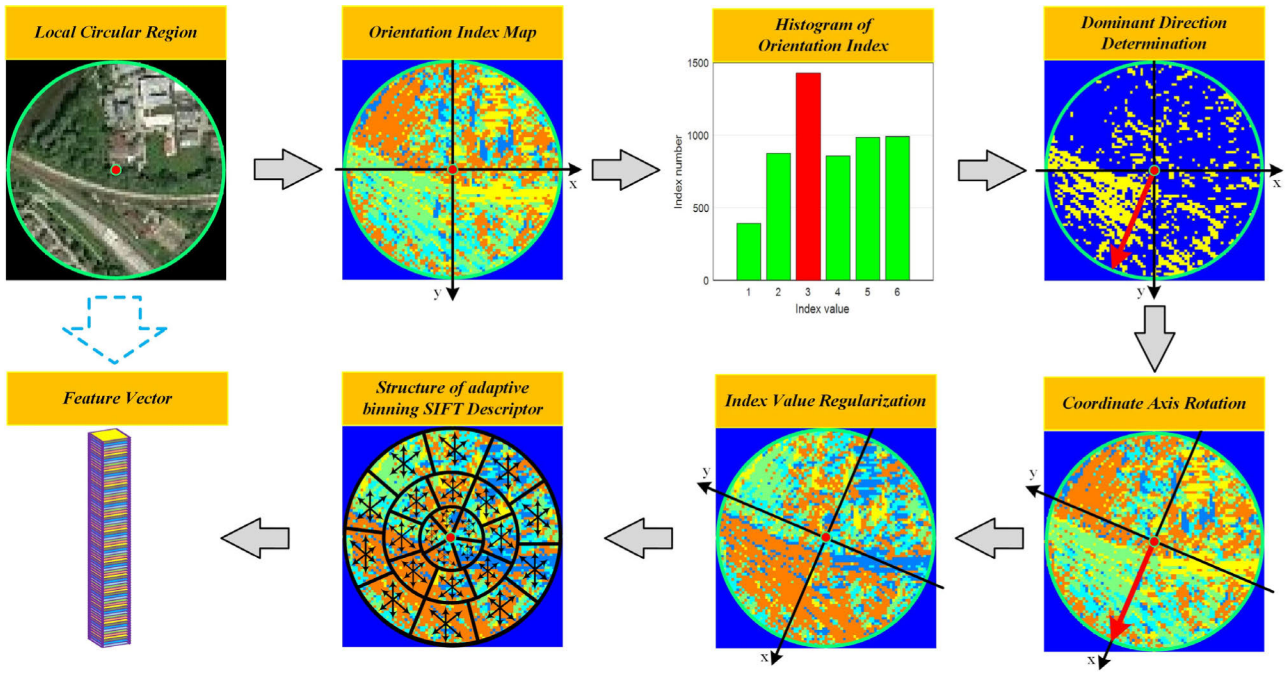


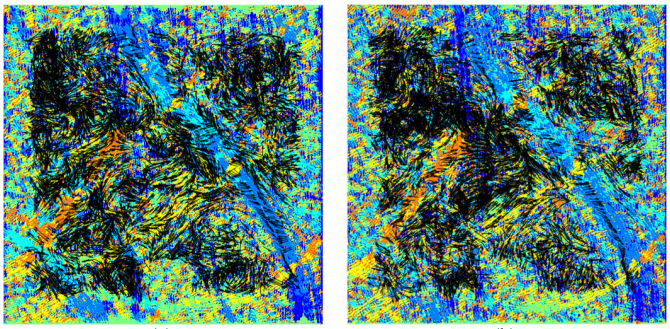
Fig. 6. Construction process of the proposed rotational invariant descriptor.

generally does not reflect the statistical image characteristics, information describing the feature point must be generated by combining all pixels within its local neighborhood. Determination of the feature point direction also requires the support of pixels in its local area, thus enabling acquisition of the necessary characteristic angle. Therefore, a histogram of the orientation index values of all pixels within the local region of the target feature point is created, as shown in Fig. 6. The index histogram has N_o bins, corresponding to the number of orientations in the filter bank. The index value $\mathrm{iv}_{\mathrm{most}}$, corresponding to the peak in the histogram, is obtained as follows:

$$\begin{cases} \mathrm{iv}_{\mathrm{most}} = \mathrm{peak}_{\mathrm{iv}}(\mathrm{niv}_{\mathrm{i}}) \\ \mathrm{niv}_i = \dfrac{1}{\mathrm{iv}_i} \displaystyle\sum_{\mathrm{CW}} \mathrm{iv}_i \\ \mathrm{iv}_i = \mathrm{OIM}(x, y), \quad (x, y) \in \mathrm{CW} \end{cases} \tag{7}$$

where $\mathrm{peak}_{\mathrm{iv}}(\cdot)$ is the function used to obtain index value $\mathrm{iv}_{\mathrm{most}}$; niv_i is the number of index value iv_i, where i represents the ith bin; and CW denotes the circular window.

The subregion corresponding to $\mathrm{iv}_{\mathrm{most}}$ is regarded as the dominant region of the circular window. Other peaks within 80% of the highest peak are also used to create dominant regions, such that feature points may have multiple description vectors. Generally, the dominant region of $\mathrm{iv}_{\mathrm{most}}$ occupies the largest proportion of the area in the image window; it is the best subregion to describe the characteristics of the entire region. Therefore, the dominant direction of the target feature point is determined based on the dominant region. Because the index value is an integer within $[1, N_o]$, the dominant region can be regarded as a sheet of constant density and thickness, and the centroid (x_C, y_C) of the dominant region



(a) (b)

Fig. 7. Results of dominant direction of feature points based on the OIMs obtained using H2StF. (a) Result of the optical image in Fig. 5(a). (b) Result of the SAR image in Fig. 5(a). The black arrows pointing to the dominant direction of the features.

can be precisely calculated, as follows:

$$\begin{cases} x_C = \dfrac{\sum x\mathrm{OIM}(x, y)}{\sum \mathrm{OIM}(x, y)}, \quad (x, y) \in \mathrm{DR} \\ y_C = \dfrac{\sum y\mathrm{OIM}(x, y)}{\sum \mathrm{OIM}(x, y)}, \quad (x, y) \in \mathrm{DR} \end{cases} \tag{8}$$

where DR denotes the dominant region. The direction of the vector connecting the feature point and (x_C, y_C) is used as the dominant direction. The angle θ_{DR}, which indicates the dominant direction, is calculated as $\theta_{\mathrm{DR}} = \arctan2(y_C/x_C)$, where $\arctan2(\cdot)$ is the four-quadrant arctangent function. Fig. 7 shows the dominant direction of each feature extracted from the image pair in Fig. 5(a).

Next, the positive x-axis of the sampling coordinate system is rotated to the dominant direction to generate the descriptor with rotational invariance. Because the $\mathrm{iv}_{\mathrm{most}}$ value corresponding to the peak of the histogram cannot be determined under different rotational angles, index value regularization is

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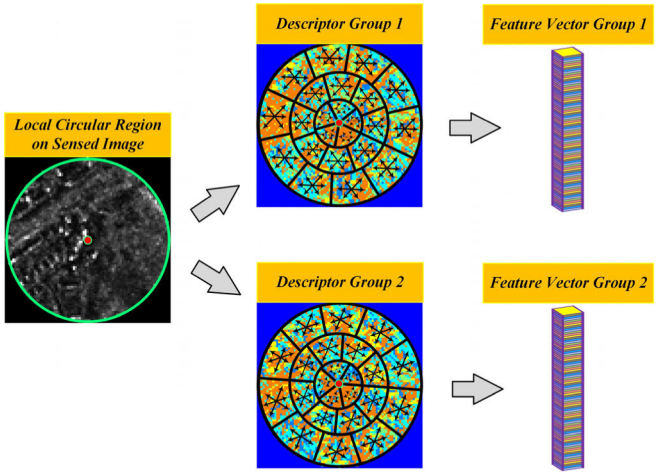


Fig. 8. Construction process of the grouped descriptors of sensed image.

performed to update the index values in the circular window. After completion of the regularization process, reasonable description vectors are generated. The regularization process is conducted in accordance with the rule of maximizing $\mathrm{iv}_{\mathrm{most}}$, which is expressed as follows:

$$\begin{cases} \mathrm{IV}_i = \mathrm{iv}_i + N_o - \mathrm{iv}_{\mathrm{most}}, & \mathrm{iv}_i \le \mathrm{iv}_{\mathrm{most}} \\ \mathrm{IV}_i = \mathrm{iv}_i - \mathrm{iv}_{\mathrm{most}}, & \mathrm{iv}_i > \mathrm{iv}_{\mathrm{most}} \end{cases} \tag{9}$$

where the IV_i is the updated index value after regularization.

Feature vectors are obtained using a distribution histogram technique and the structure of adaptive binning SIFT descriptor [35]. According to Fig. 6, the circular window is first separated into three nonoverlapping rings, which are further uniformly divided into 5, 8, and 10 subregions from the inside out, respectively. Then, a local distribution histogram of the orientation index for each subregion is created. After that, the feature vector is generated by the ordered concatenation of these histograms. Finally, the L2-norm is used to normalize the feature vector to output the final result.

However, extensive experiments revealed that the performance of the proposed descriptor did not meet our expectations because of orientation sampling sparsity when the OIM was constructed. The number of orientations N_o is generally set to small even numbers (e.g., 6, 8, or 10). For example, if N_o is set to six, the angle difference between two adjacent orientations o_i and o_{i+1} is equal to $\pi/6$. The similarity of OIMs between reference and sensed images will reach a minimum at angle differences near odd integer multiples of $\pi/12$. In this context, the number of correct matches (NCM) also reaches a minimum. However, N_o cannot be set too large because this would lead to huge computational costs in steps such as convolutional calculation and feature retrieving. To address this problem, a descriptor-grouping strategy was used for the sensed image, with the aim of further improving the rotational invariance of OFM. Specifically, the relationship between the orientation numbers of the two images is $N_{o,\mathrm{sen}} = 2 \cdot N_{o,\mathrm{ref}}$, where the subscripts sen and ref indicate sensed and reference images, respectively. Two OIMs were constructed for the sensed image by dividing the convolution results into two groups, $\{R_{o_1}, R_{o_3}, \ldots, R_{N_{o,\mathrm{sen}}-1}\}$ and $\{R_{o_2}, R_{o_4}, \ldots, R_{N_{o,\mathrm{sen}}}\}$; this yielded two groups of descriptors. The construction of these grouped descriptors for the sensed image is shown in Fig. 8.

C. Feature Matching and Mismatch Elimination

After the description vectors of all extracted features are calculated, feature matching is performed. Initial matches are obtained through a brute-force matching method [36], and cross-checking is conducted to return consistent pairs through dual matching. Although this method maximizes the reliability of the initial match set, many mismatches remain within the results, particularly when matching multisource, multitemporal remote sensing images. The inlier rate of initial matches is usually low because of significant NIDs between images; this low inlier rate causes further instability in conventional mismatch removal methods and leads to a need for more iterations to identify optimal putative matches [9]. Therefore, a mismatch elimination method (comprising a RANSAC algorithm and MSIO strategy) is implemented to ensure the accuracy and reliability of matching results.

A transformation model is required to build the framework of the RANSAC. In this study, we used a global affine model to represent the relationship between the reference and sensed images. In our framework, the number of iterations in RANSAC is typically only 1500, such that the consensus set is presumably suboptimal. Therefore, the MSIO strategy is designed to further optimize the results and obtain the final putative matches. The MSIO input data is the consensus set output by RANSAC. First, based on the least-squares rule, the entire consensus set is used to estimate an affine model. Then, the transformation errors of all matches in the outlier set are calculated; matches with errors below a specific threshold are moved into the consensus set. Next, the affine model is reestimated using the consensus set, where matches with errors greater than three standard deviations are moved into the outlier set. This process is repeated until no matches in the outlier set are moved into consensus set, and the MSIO process is considered complete. The process of the proposed MSIO strategy is provided in Algorithm 1.

Generally, if the consensus set is not excessively far from the optimal putative match set, the MSIO process will end after approximately five iterations. If higher matching accuracy is necessary, a coarse-to-fine matching strategy can be used to repeat image matching after the elimination of significant geometric differences between the images. However, if the output consensus set is nearly filled with mismatches, MSIO will become stuck in an endless loop. To avoid this outcome, the number of iterations n is recorded and the loop is forced to break if n exceeds a specific number.

IV. Experiments and Analysis

In this section, the performance of the proposed method was investigated using a large-scale dataset that contained 4427 pairs of multisource, multitemporal images. The performance of the proposed method was compared with the performances of five state-of-the-art methods: LGHD; RIFT; OSS; ASS; and LNIFT, all of which were implemented

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Algorithm 1 Match Set Iterative Optimization (MSIO)

Input:
- M_{ini}: the initial match set;
- C: the consensus set;
- $Thred$: the threshold to reject mismatches.

Output: the putative match set P

1) Obtain the outlier set O by moving C from M_{ini}.
2) $n = 0$.
3) **While** 1
4) $n = n + 1$.
5) Estimate the affine transformation model A with C and move the matches with transformation error less than $Thred$ in O into C.
6) **if** no match in O is moved into C **or** $n > 100$, **do**
7) $P = C$.
8) **break**
9) **end if**
10) Estimate the affine transformation model A with C again and move the matches with transformation error greater three times the standard deviation in C into O.
11) **end** while

using publicly available code with optimal parameters settings provided by their authors.

A. Datasets

Seven real remote sensing image datasets were collected, including two multitemporal optical image datasets and five multisource image datasets. The source datasets used to construct our datasets were Mts-WH [37], LEVIR-CD [38], WHU-OPT-SAR [39], OS [40], LLVIP [41], DIML/CVL [42], and MCDDS [43], which are mainly used as training and test datasets for deep learning tasks. Because the image pairs in these datasets are preregistered, their ground truth transformation matrices are all identity matrices, thus providing a reference for quantitative evaluation of image-matching results. To increase the diversity of the experimental data, these preregistered images were artificially subjected to random translations (−30 to 30 pixels in the row and column directions), scale ratios (1:1–1:3), and rotational angles (0°–360°); their ground truth transformation matrices were then recalculated. Typical image pairs in the recreated datasets are shown in Fig. 9, and the datasets are described in detail in Table I.

B. Evaluation Criteria and Parameter Settings

In this experiment, six criteria were used to evaluate the matching performance of our OFM method: NCM, precision, recall, inlier ratio, root mean square error (RMSE), and success rate (SR).

To define correct matches, putative matches were evaluated by the ground truth transformation model of the image pair. Matches with location errors of <3 pixels were regarded as correct matches. Precision was calculated as NCM/NP, where NP is the number of putative matches; recall was calculated as NCM/NC, where NC is the number of correspondences.

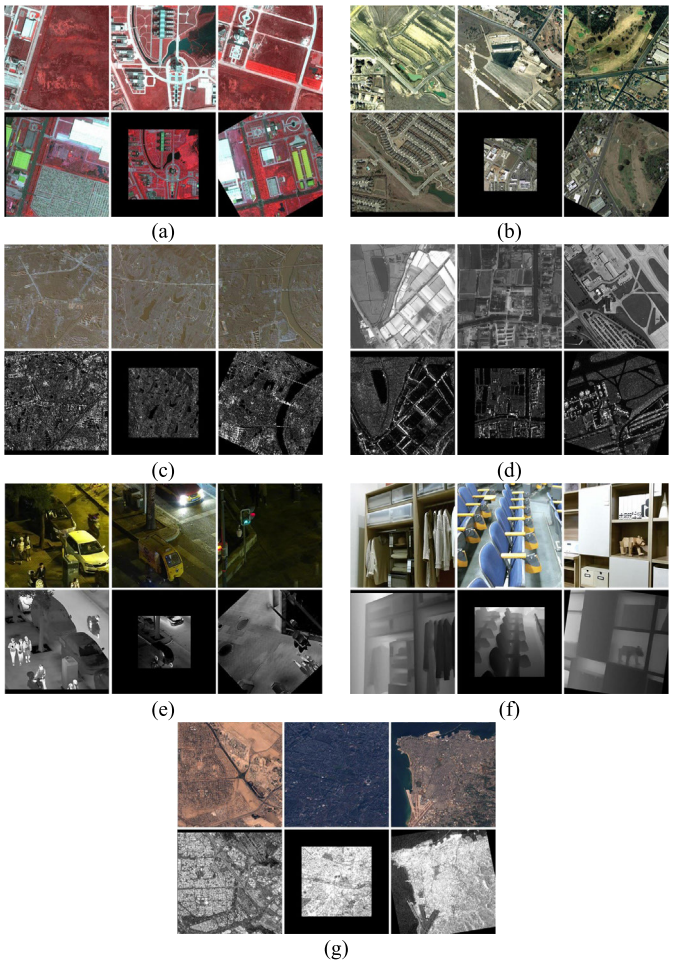


Fig. 9. Some typical image pairs of the created image datasets. (a) Dataset 1: multitemporal optical–optical images. (b) Dataset 2: multitemporal optical–optical images. (c) Dataset 3: optical–SAR images. (d) Dataset 4: optical–SAR images. (e) Dataset 5: optical–infrared images. (f) Dataset 6: optical–depth images. (g) Dataset 7: optical–SAR images.

TABLE I
DETAILS OF EXPERIMENTAL DATASETS

No.	Category	Quantity (pairs)	Image Source	GSD	Size (pixels)
1	Optical-to-Optical	83	IKONOS	1m	600×600
2	Optical-to-Optical	425	Google Earth	0.8m	641×641
3	Optical-to-SAR	1899	Gaofen-1 Gaofen-3	5m	600×600
4	Optical-to-SAR	985	Google Earth Gaofen-3	1m	512×512
5	Optical-to-Infrared	898	HIKVISION DS-2TD8166	centimeter -level	683×683
6	Optical-to-Depth	203	Microsoft Kinect v2	centimeter -level	756×756
7	Optical-to-SAR	24	Sentinel-2 Sentinel-1	10m	600×600

The inlier ratio was calculated as NCM/NIM, where NIM is the number of initial matches. RMSE reflects the accuracy of the estimated transformation model generated by the putative matches; to calculate the RMSE for each image pair, 100 uniformly distributed checkpoint pairs were generated using the ground truth transformation model. Lower RMSE

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TABLE II
PARAMETERS SETTINGS OF OFM

Method	Parameter settings
OFM-G1StF	$r = 42$, $N_{o,ref} = 6$, $N_s = 4$ (σ_1=1, σ_2=2, σ_3=3 and σ_4=4)
OFM-G2StF	$r = 42$, $N_{o,ref} = 6$, $N_s = 4$ (σ_1=1, σ_2=2, σ_3=3 and σ_4=4)
OFM-H2StF	$r = 42$, $N_{o,ref} = 6$, $N_s = 4$ (σ_1=2, σ_2=4, σ_3=6 and σ_4=8)
OFM-GoF	$r = 42$, $N_{o,ref} = 6$, $N_s = 4$ (f_1=0.4, f_2=0.28, f_3=0.2 and f_4=0.14)
OFM-LGoF	$r = 42$, $N_{o,ref} = 6$, $N_s = 4$ (f_1=0.33, f_2=0.21, f_3=0.13 and f_4=0.08)

is associated with higher accuracy; thus, successfully matched image pairs were pairs for which the RMSE was <3 pixels. SR was calculated as NSM/N, where NSM is the number of successfully matched image pairs in the dataset and N is the number of image pairs in the dataset.

There are three key parameters in our OFM method: the number of orientations of the filter bank used on reference image $N_{o,\text{ref}}$, the number of scales of filter bank N_s, and the radius of feature descriptor r. These parameters were tested using the datasets, as shown in Table I, and the recommended optimal settings were as follows: $N_{o,\text{ref}} = 6$, $N_s = 4$, and $r = 42$. The orientation angles of the different filters were identical, but the scale factors differed among filters. The scale factor was set to the Gaussian standard deviation σ for Gaussian steerable filters; it was set to the center frequency f for Gabor and log-Gabor filters. The detailed parameters settings of OFM are listed in Table II. The number of iterations and residual threshold of RANSAC were set to 1500 and 3 pixels, respectively. The Gaussian-scale space was set to five layers, with scale factors of 1, 1.5, 2, 2.5, and 3; the maximum number of extracted feature points was 5000, which matched the maximum numbers for the compared algorithms.

C. Evaluation of Rotational Invariance

To evaluate the rotational invariance of the proposed OFM algorithm, experiments were conducted using the optical–SAR image pair shown in Fig. 5(a). Because the two images had almost no rotational difference, the SAR image was artificially rotated 72 times in 5° steps to generate rotational differences from 0° to 360°. The results are expressed in terms of the NCM per rotational angle. The experimental results are compared in Fig. 10. For better visualization, only the results of our OFM framework are plotted for only G1StF and GoF.

The results shown in Fig. 10 demonstrate that RIFT, OSS, and ASS achieved rotational invariance. However, the RIFT and ASS results were very unstable. Although the OSS results are relatively stable, it has an overall low NCM under different rotational angles. LGHD and LNIFT performed poorly, with their results substantially decreased after the rotational angles reached 15° and 110°, respectively. Among all tested approaches, OFM showed the best performance. This experiment briefly illustrates the rotational invariance of OFM; additional experimental results will be reported in the next subsection.

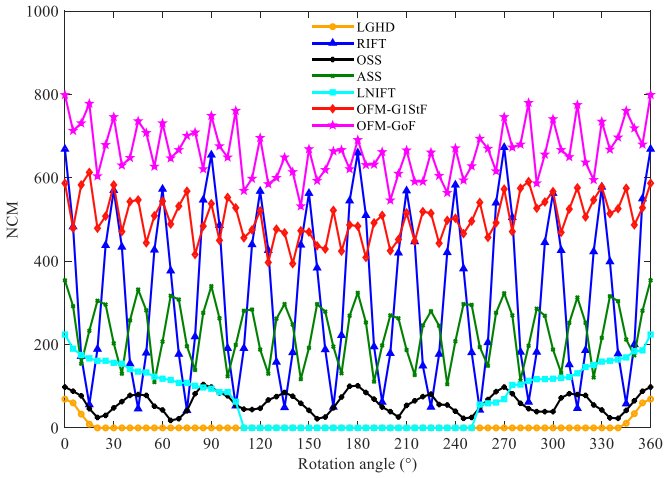


Fig. 10. Matching performance of LGHD, RIFT, OSS, ASS, LNIFT, and OFM under different rotation angles.

D. Qualitative and Quantitative Results

For qualitative evaluation, multiple typical image pairs were selected from each dataset; the correct matches determined by using each method on these image pairs were visualized. Automatic and precise matching of these image pairs is extremely challenging because of their significant NIDs and geometric differences. The qualitative comparison results for LGHD, RIFT, OSS, ASS, LNIFT, and our OFM are presented in Figs. 11–13.

The results in Figs. 11–13 indicate that LGHD yielded some correct matches on image pairs with translational and scale differences, but it completely failed on image pairs with rotational differences. These results demonstrate that LGHD is sensitive to image scale and rotation variation, and it cannot resist large geometric differences. RIFT, OSS, ASS, and LNIFT showed similar performance, yielding some correct matches on most image pairs. These findings indicate that these methods are resistant to NIDs between images. OSS and ASS performed better than RIFT and LNIFT on images with scale differences. However, they obtained comparatively few correct matches and exhibited a nonuniform distribution of points, particularly on images with rotational differences. In contrast, the methods in our OFM framework (OFM-G1StF, OFM-G2StF, OFM-H2StF, OFM-GoF, and OFM-LGoF) achieved the best performance overall. The results of various multioriented filters differed; the matching results of G1StF, H2StF, and LGoF were better than the matching results of G2StF and GoF. Each filter exhibited distinct performance according to the types of image pairs. For example, G2StF and GoF were suboptimal for the matching of optical and SAR image pairs.

To ensure that comparisons among methods were fair and quantitative, only the results of successfully matched image pairs were recorded. The results of applying the six metrics (NCM, precision, recall, inlier ratio, RMSE, and SR) to all datasets are shown in Fig. 14. These results indicate that the proposed OFM method outperformed all other methods for nearly all datasets, thereby demonstrating the effectiveness and advantages of the multiscale feature detection strategy, pooling-based feature dimensionality reduction operation,

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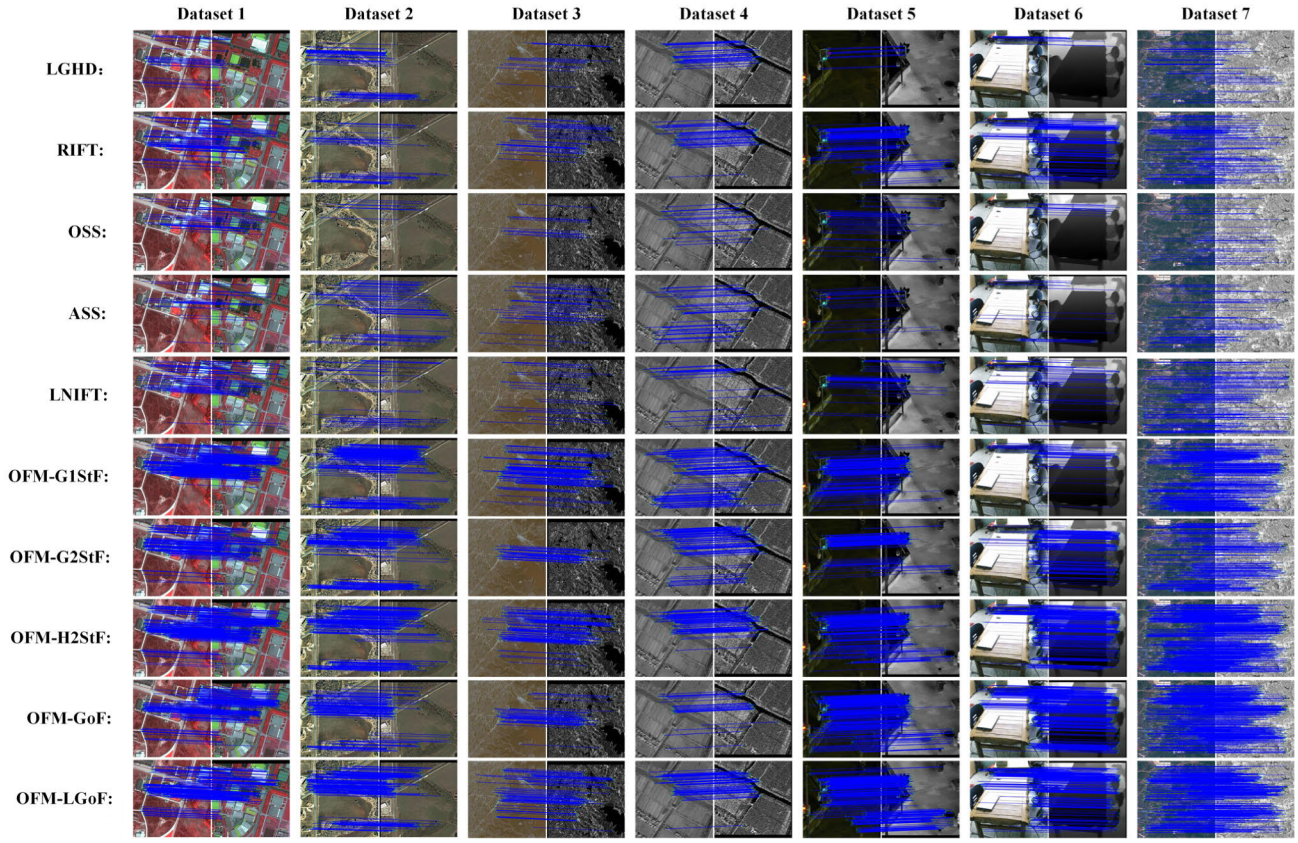


Fig. 11. Correct matches of each method on some typical image pairs with translational differences. Blue lines represent correct matches.

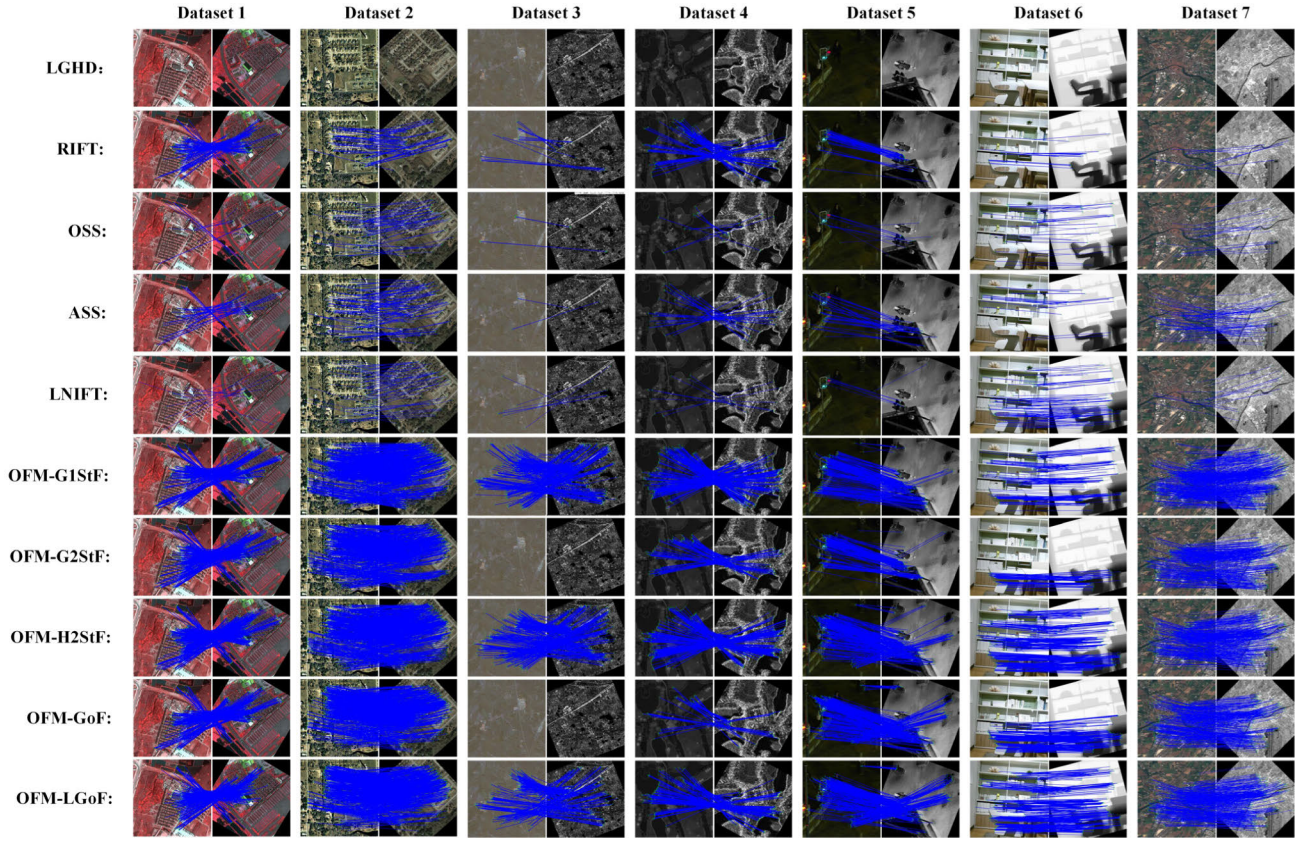


Fig. 12. Correct matches of each method on some typical image pairs with rotational differences. Blue lines represent correct matches.

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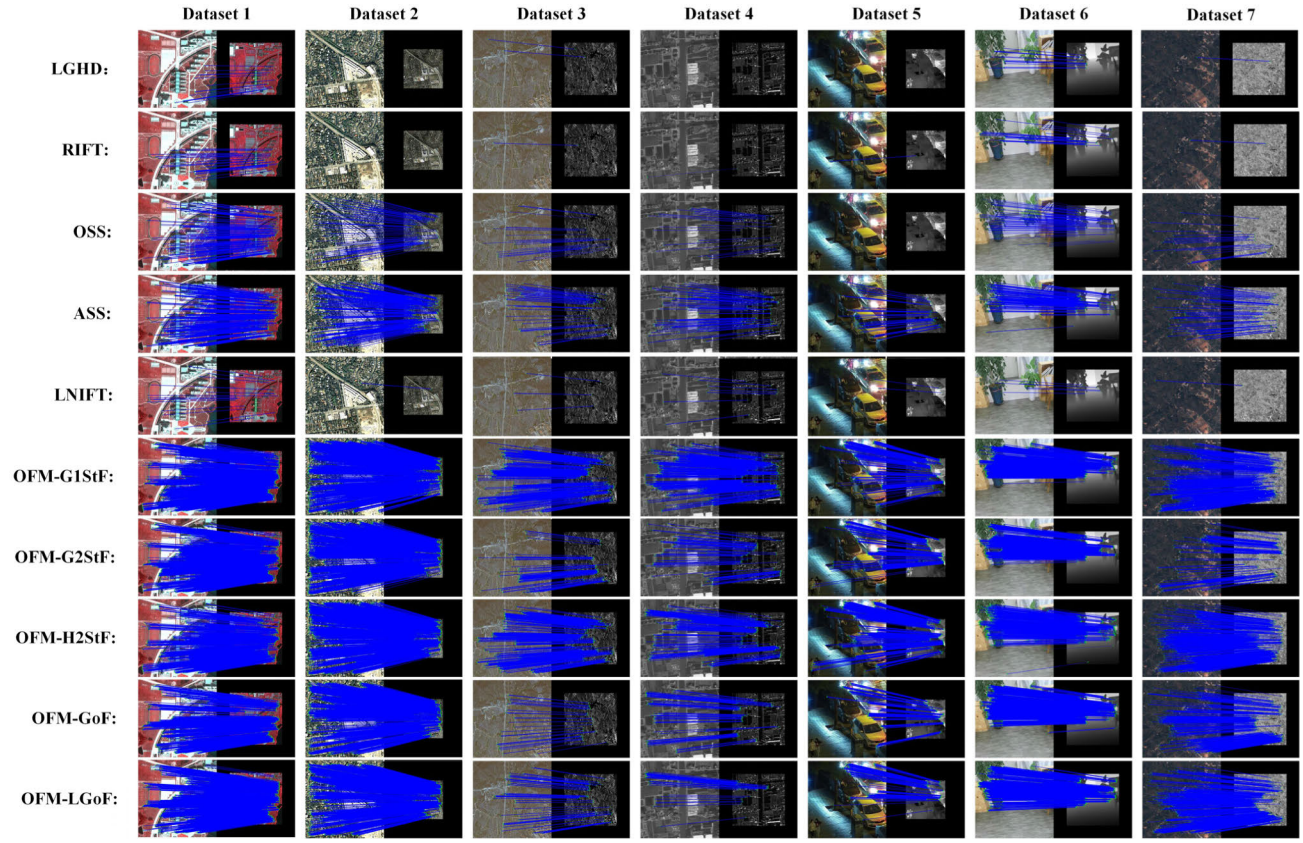


Fig. 13. Correct matches of each method on some typical image pairs with scale differences. Blue lines represent correct matches.

novel dominant direction estimation approach, descriptor-grouping strategy, and mismatch elimination method.

According to the results presented in Fig. 14, LGHD exhibited the worst performance for nearly all datasets, indicating that it is highly sensitive to NIDs between images. The NCM results of the LGHD method exceeded 100 only for dataset 2; its precision, recall, and SR results were low for all datasets. The OSS outperformed the LGHD only, and it had the lowest inlier ratio among all methods. RIFT, ASS, and LNIFT showed very similar results across all datasets, with the exception of the inlier ratio, which were significantly lower for ASS and LNIFT than for RIFT in all datasets. An overall comparison of the quantitative results is presented in Table III, including 4427 image pairs among all datasets.

Table III shows that the proposed OFM methods achieved the best results; only its precision results were slightly lower than the precision results for RIFT. Importantly, SR (with RMSE as a precondition) was the metric that most clearly reflected the overall performance of these matching methods, such that higher SRs were associated with greater stability in application. The SR results of LGHD, RIFT, OSS, ASS, and LNIFT were lower than 40% for all datasets; the proposed OFM-G1StF achieved an SR of 83.72%, which was the highest among all methods. The SR of OFM-G2StF was 47.53%, which was the lowest among methods in OFM framework but remained significantly better than the compared methods.

The OFM methods also outperformed the other methods in terms of NCM, recall, inlier ratio, and RMSE. Among these, the NCM and inlier ratio indicated the highest OFM performance. NCM exceeded 230 for all OFM methods, whereas the best compared method (LNIFT) exhibited NCM < 140. The inlier ratio for OSS was only 1.29%, whereas the inlier ratios for all OFM methods were $>10.7\%$. These comparison results demonstrate the robustness, effectiveness, and superiority of OFM.

E. Analysis Results

The qualitative and quantitative results indicate that LGHD can be used to match images with NIDs but is inappropriate for use in matching images with large geometric differences; its poor resistance to geometric differences and low matching SR severely limit its practical application scenarios. Compared with LGHD, RIFT, OSS, and ASS performed better for each dataset, but they also showed low SRs, particularly on optical–SAR and optical–infrared datasets, which have severe NIDs. LNIFT performed slightly better, but it still had low results on most metrics. Although RIFT showed the highest mean precision on its correctly matched image pairs, it performed poorly according to other evaluation metrics. In contrast, our OFM showed high robustness and generality for various types of datasets, yielding the highest NCM and SR with the highest accuracy.

There are two main reasons for the poor performance of LGHD on our dataset. First, the feature reduction of its descriptors is incomplete. LGHD uses multiscale, multioriented log-Gabor filters to construct its index map, but

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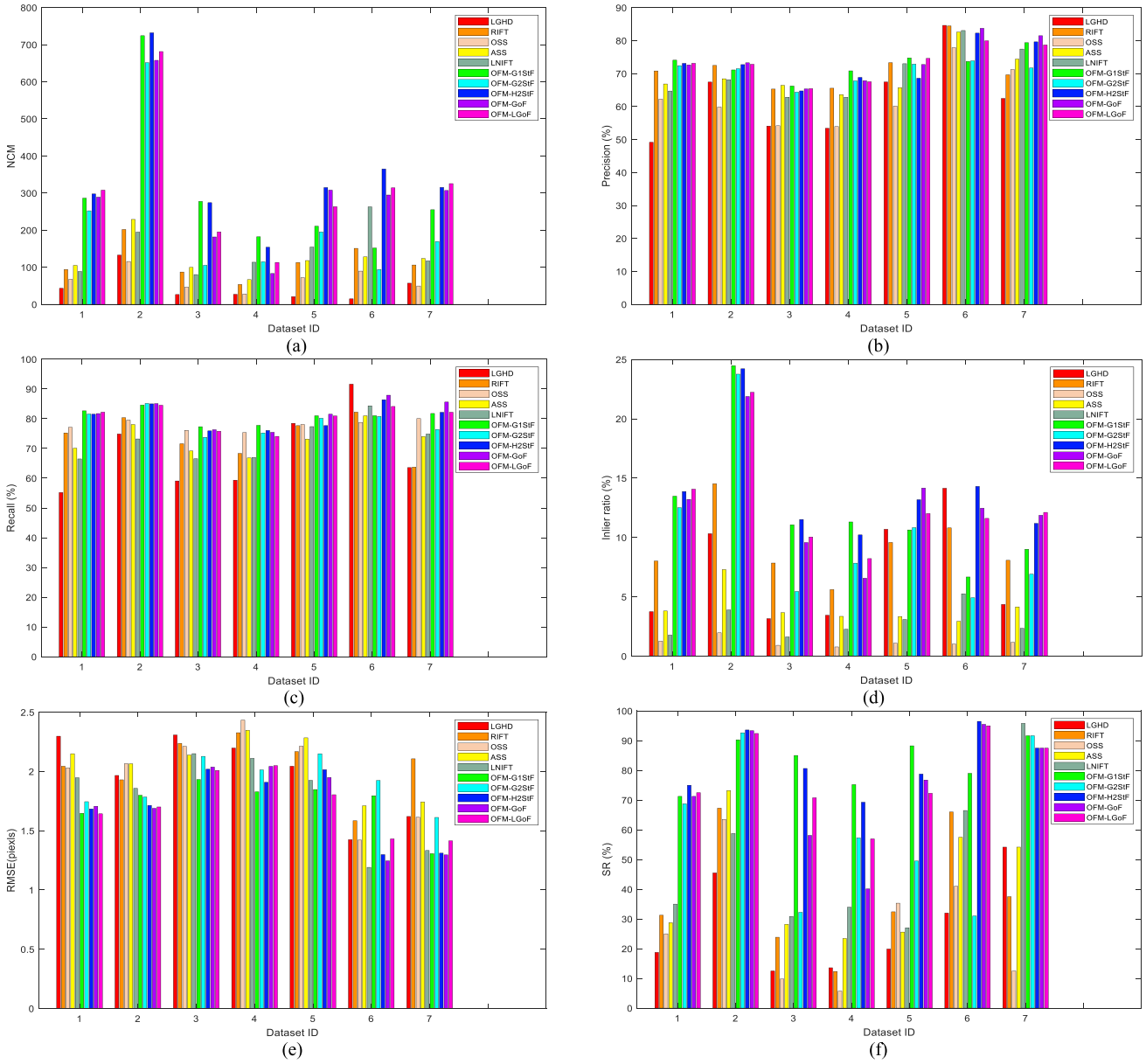


Fig. 14. Quantitative comparison of different metrics for various datasets. (a) NCM results. (b) Precision results. (c) Recall results. (d) Inlier ratio results. (e) RMSE results. (f) SR results.

TABLE III
AVERAGE RESULTS ON ALL DATASETS (4427 IMAGE PAIRS)

Method	Evaluation criteria					
	NCM (pairs)	Precision (%)	Recall (%)	Inlier ratio (%)	RMSE (pixels)	SR (%)
LGHD	50.22	62.44	69.56	7.41	2.07	18.59
RIFT	121.91	**70.75**	75.67	9.81	2.09	29.53
OSS	78.64	60.17	78.01	1.29	2.11	21.01
ASS	128.37	67.72	72.38	4.30	2.14	32.51
LNIFT	132.75	67.22	71.01	2.66	1.96	35.43
OFM-G1StF	**286.68**	69.92	**79.21**	**12.24**	**1.86**	**83.72**
OFM-G2StF	**232.83**	68.91	**78.04**	**10.78**	2.01	**47.53**
OFM-H2StF	**318.24**	68.33	**78.03**	**13.24**	**1.91**	**79.86**
OFM-GoF	**277.69**	70.11	**79.68**	**12.33**	**1.90**	**63.73**
OFM-LGoF	**266.23**	69.67	**78.32**	**11.84**	**1.88**	**71.58**

it does not eliminate the scale dimension of the feature maps. This strategy leads to the retention of a large number of invalid features, while significantly increasing the computational effort. Second, LGHD does not have scale and rotational invariance. Numerous image pairs in our datasets have scale and rotation variation; LGHD was rarely able

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to match these images. RIFT also uses log-Gabor filters to construct index maps, and it achieves good feature reduction by summing the feature maps. RIFT uses an iterative approach to sequentially modify the starting layer of the feature map to imitate the rotation of the image, thus achieving resistance to rotational differences between images. However, it only performs well for specific rotation angles between images, and it is highly unstable when the rotational angle is unpredictable. Our experimental results suggest that the log-Gabor filter is not the optimal multioriented filter for the construction of index maps. When RIFT was applied to optical–SAR image pairs, its performance sharply decreased. ASS constructs its index map based on local self-similarity descriptors and determines the dominant direction by sequential rotation of the descriptors. Compared with the RIFT strategy of rotating the whole image, ASS exhibits better stability for different image rotational angles. However, ASS could not manage large NIDs; therefore, it performed poorly on the optical–SAR and optical–infrared datasets. Because the local self-similarity descriptor has no directionality, numerous useless descriptors were generated, leading to the low inlier ratio results for ASS. OSS employed an offset mean filtering method to obtain the improved self-similarity features, which were used to establish a multichannel self-similarity map. OSS achieves scale invariance by constructing a multiscale pyramid of the feature maps. And the self-similarity values around the target feature points are used to generate histograms for the estimation of dominant directions. However, OSS uses only 36 points (with an interval of 10° and covering 360°) for dominant direction estimation, which leads to low accuracy of the results. As with ASS, despite the use of improved self-similarity features, OSS generates a large number of invalid descriptors, which reduces the inlier ratio of matching results. LNIFT uses a locally normalization filter based on an average filter to convert multisource images into the similar modalities. Therefore, the multisource matching problem becomes a homo-source image matching problem. An improved ORB detector is used to extract feature points on the normalized images while an adaptive nonmaximal suppression strategy is employed. A simplified HOG (using a four-bin histogram) descriptor is used for feature description. However, according to the results presented in Fig. 10, LNIFT can only handle image pairs with rotational differences between $-90°$ to 90°. This is because LNIFT normalizes the angle from [0°, 360°] to [0°, 180°] when calculating the dominant direction. This drawback will limit its practical application scenarios.

Unlike the other methods, OFM uses five different multioriented filters (G1StF, G2StF, H2StF, GoF, and LGoF) to construct index maps that can be used for the exploration of their performances among different types of datasets. Based on the multidirectional convolution results, OFM first reduces the dimensionality of the feature maps using pooling techniques to obtain index maps that are robust to NIDs. Then, it uses a novel approach (based on the centroid of the dominant region) for stable estimation of the dominant directions of the descriptors. An effective descriptor-grouping strategy enhances the resistance of the method to image rotational changes. The mismatch elimination algorithm, which combines RANSAC and MSIO strategies, ensures more correct matches in the putative match set. The performances of the different filters in the OFM framework exhibited considerable variation: G1StF and H2StF performed best, followed by GoF, LGoF, and G2StF. The main differences among filters are that G1StF and H2StF are odd-symmetric filters, G2StF is an even-symmetric filter, and GoF and LGoF are filters with both odd- and even-symmetric parts. Both odd- and even-symmetric filters were clearly resistant to NIDs, but odd-symmetric filters were more resistant to large NIDs. The experimental results revealed that the performances of GoF and LGoF filters with both odd- and even-symmetric parts were worse than the performances of G1StF and H2StF. This phenomenon may have occurred because even-symmetric filters can only extract information in the direction of the kernel function, whereas odd-symmetric filters can extract information in two directions (the direction of the kernel function and the vertical direction). During image convolution, even-symmetric filters add values on both sides of their kernel functions, whereas odd-symmetric filters subtract the values on both sides of their kernel functions. This difference may lead to better differentiation of the convolution results by the odd-symmetric filter in distinct orientations, resulting in better matching performance.

F. Discussion of Related Remote Sensing Applications

Different sensors provide complementary information about earth observations, and many applications would benefit from the joint use of multisource images. Accurate and stable image matching algorithms are essential for subsequent applications based on multisource images, such as multisource change detection and multisource image fusion. On the one hand, the same ground area will be aligned better with the support of accurate co-registration. This will be helpful in reducing the unfavorable factors in the process of change detection and image fusion, and their results will be more reliable. On the other hand, stable matching algorithms can work in different ground cover scenarios, such as farmland, mountains, and water areas, which are of great importance for monitoring natural resources with change detection methods.

V. Conclusion

In this study, we proposed a novel multisource, multitemporal remote sensing image matching method OFM based on multioriented filters. The OFM method consists of the following steps: multiscale feature extraction, index map construction based on multidirectional convolutions, rotationally invariant descriptor computation based on the centroid of the local dominant region, and mismatch elimination using a combined RANSAC algorithm and MSIO strategy. The OFM method is invariant to scale and rotational changes, and it is robust to significant NIDs. Five multioriented filters (G1StF, G2StF, H2StF, GoF, and LGoF) applicable to the OFM framework were explored in extensive experiments using 4427 image pairs with large geometric differences and NIDs. The results demonstrated that the OFM method outperformed the state-of-the-art methods LGHD, RIFT, and ASS in terms of NCM, recall, inlier ratio, RMSE, and SR.

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However, OFM is not invariant to affine deformations. Future work will focus on making OFM methods affine invariant. To achieve this goal, there is a need to consider extracting feature points with affine invariance, potentially using an autocorrelation matrix to iteratively estimate the affine invariant region. Furthermore, the shape of the feature descriptors should be adaptively adjusted according to the extracted affine invariant regions, ensuring the validity of the description information. Finally, deep learning techniques may be adapted to optimize the index map structure, thereby improving the matching effect.

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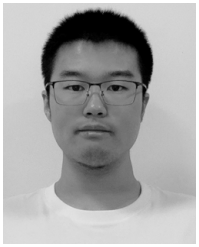
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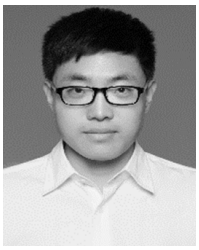
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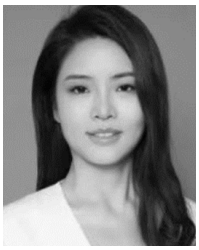
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